Lecture 5: AWS Storage

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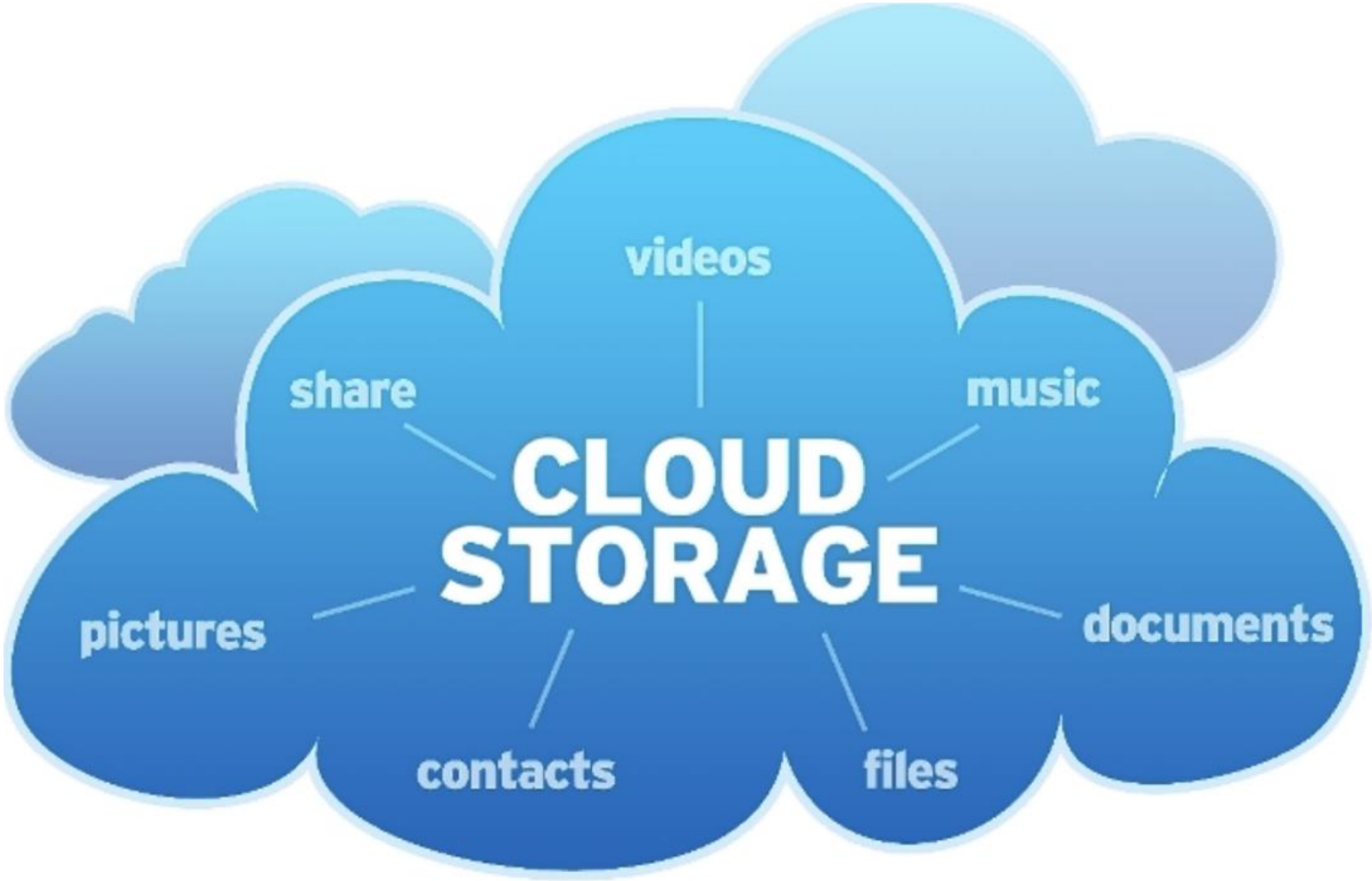
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Agenda

- **Introduction to Cloud Storage**
- **Overview of AWS Storage Services**
- **Amazon S3 (Simple Storage Service)**
- **CloudFront**
- **Amazon EBS (Elastic Block Store)**
- **Amazon EFS (Elastic File System)**

Introduction to Cloud Storage



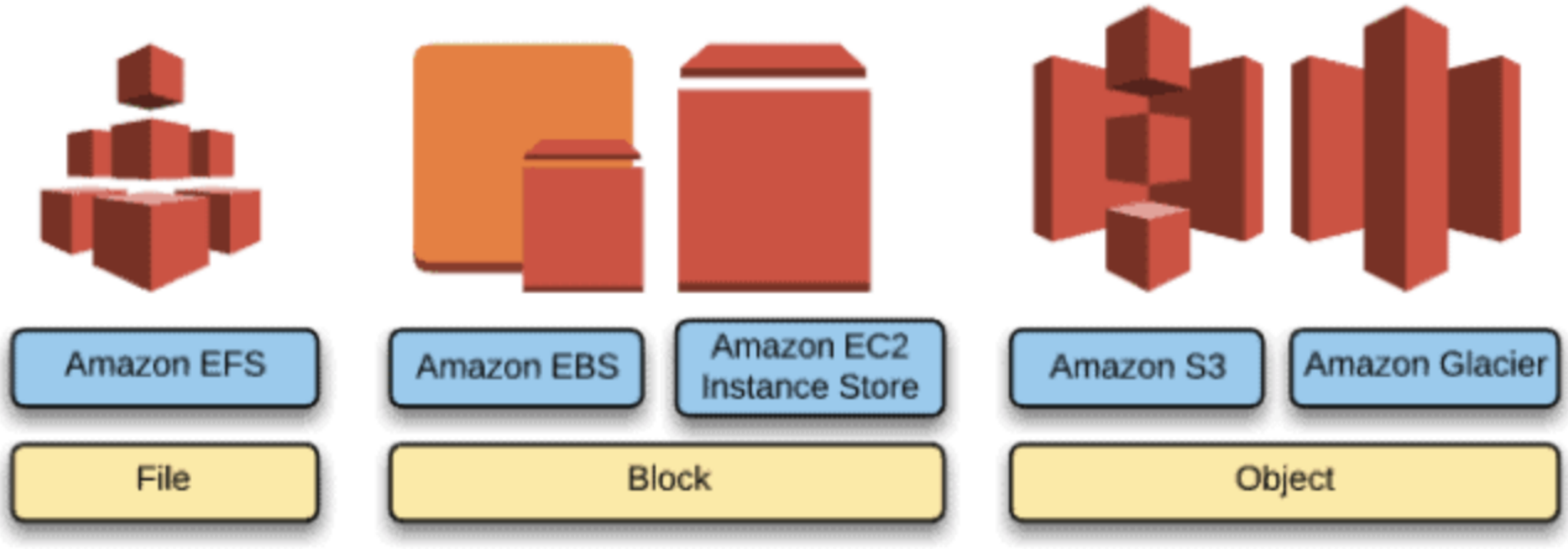
Introduction to Cloud Storage

- Store digital data across multiple virtual servers hosted by third parties
- Scalability
- Cost efficiency
- Accessibility and Availability
- Security
- Disaster recovery

Introduction to Cloud Storage

- Types of Storage
  - Object: Data is managed as objects such as photos, videos.
  - Block: Data is divided into blocks such as database.
  - File: Data is saved as files in a file system.

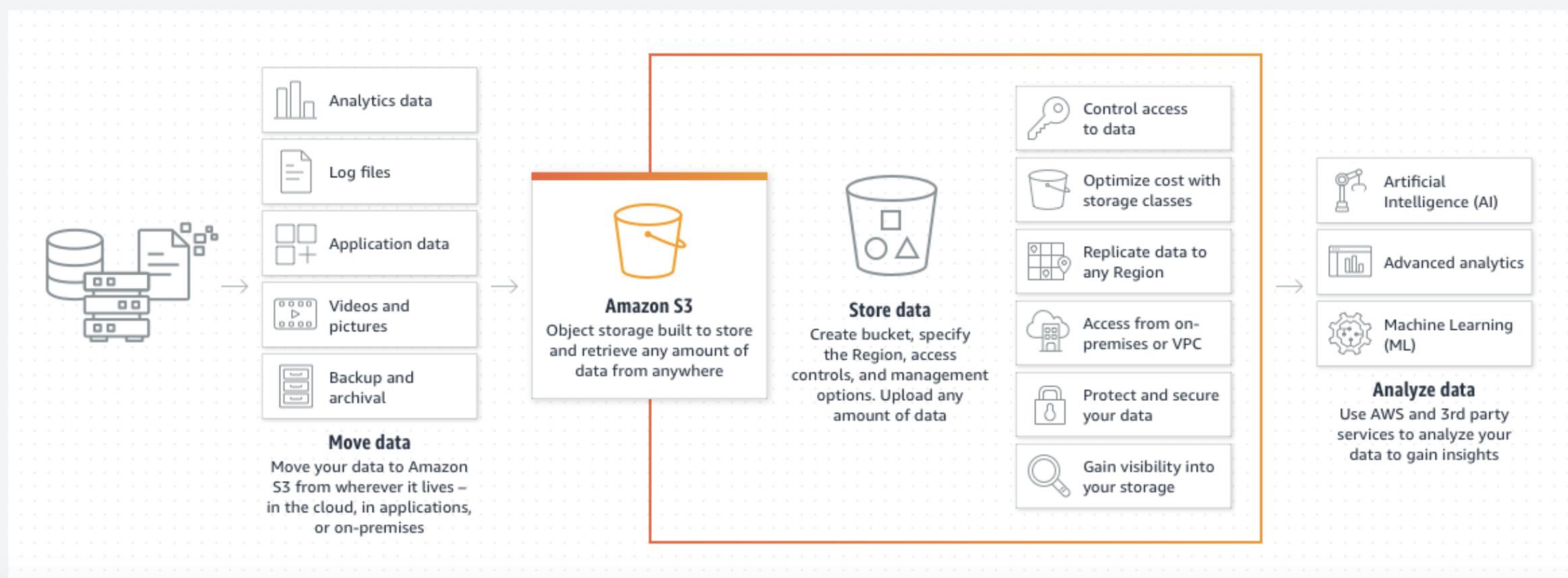
AWS Storage

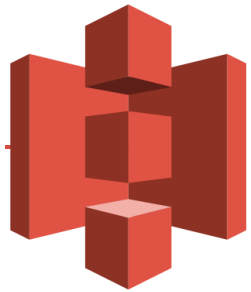

Amazon EFS
Amazon EBS
Amazon EC2
Instance Store
Amazon S3
Amazon Glacier
File
Block
Object

AWS Storage

- AWS Storage is a service that provides cloud storage
- Storage Services:
  - S3 (Simple Storage Service): is an object storage service
  - EBS (Elastic Block Store): A block-level storage service that is used with EC2. It is suitable for applications that require a file system or direct access to raw block-level storage like databases, enterprise applications
  - EFS (Elastic File System): A managed NFC (Network File System) that can be mounted on many EC2 simultaneously, suitable for applications that require a shared file system.
  - Amazon Glacier: It is for long-term archival storage, emphasizing its cost-effectiveness and security features for data archiving and backup, with retrieval times ranging from minutes to hours.

Amazon S3

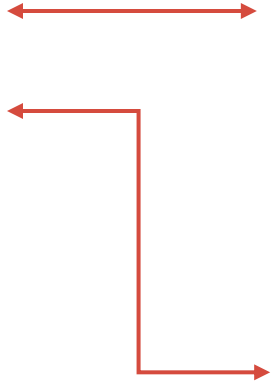

Analytics data
Log files
Application data
Videos and pictures
Backup and archival
Move data
Move your data to Amazon S3 from wherever it lives – in the cloud, in applications, or on-premises
Amazon S3
Object storage built to store and retrieve any amount of data from anywhere
Store data
Create bucket, specify the Region, access controls, and management options. Upload any amount of data
Control access to data
Optimize cost with storage classes
Replicate data to any Region
Access from on-premises or VPC
Protect and secure your data
Gain visibility into your storage
Artificial Intelligence (AI)
Advanced analytics
Machine Learning (ML)
Analyze data
Use AWS and 3rd party services to analyze your data to gain insights



AWS S3 *AWS Region (U.S. Standard)*

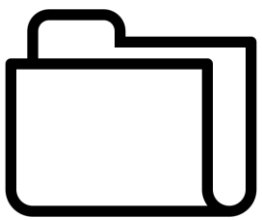


Bucket





Object



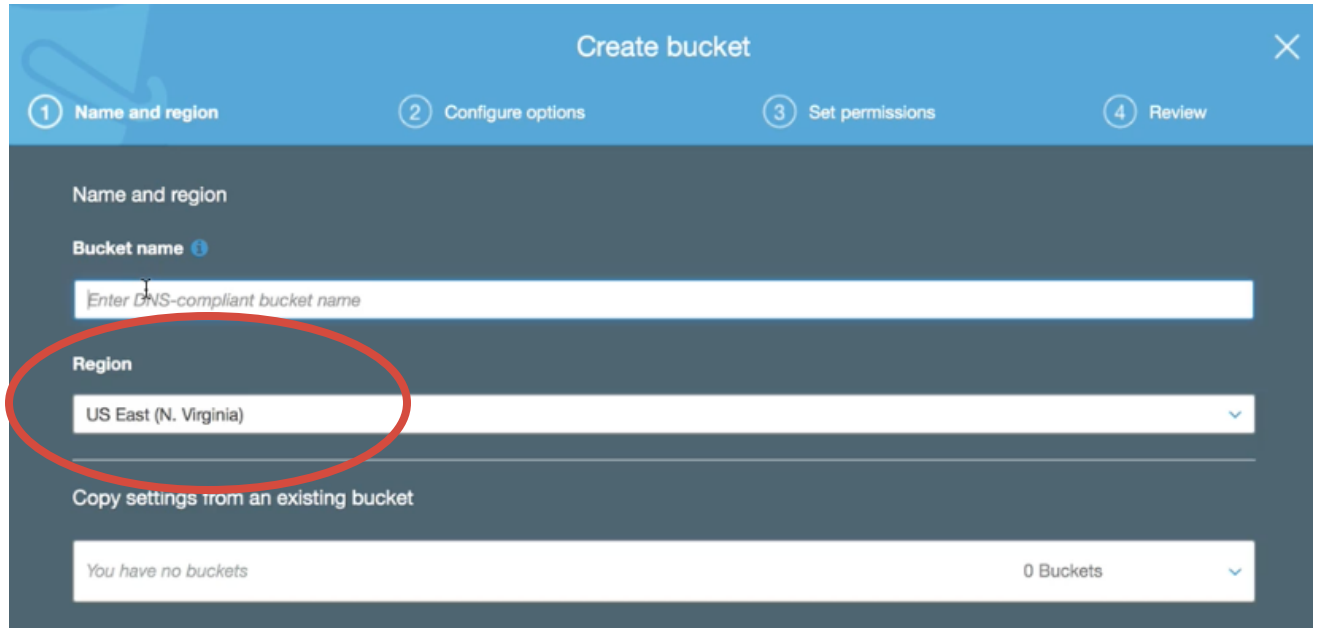
Folder



Object

AWS S3 Regions

When you create a bucket, you must select a specific region for it. This means that any data you upload to the S3 bucket will be physically located in a data center in that region. Later on, you can replicate your data in different regions if need to scale globally.



Best practice is to select the region that is physically closest to you, to reduce transfer latency. You can also stack up CloudFront on top of the S3 to reduce latency and save costs. Buckets must have a **unique name** among all AWS accounts e.g., “develop” is already take by some other AWS users.

S3 Permissions

S3 permissions allow you to have granular control over who can view, access, and use specific buckets and objects.

Permissions functionality can be found on the **bucket** and **object** levels.

There are 3 types of permissions in S3:

- **Identity-based** – IAM user or role has policies to access S3.
- **Resource-based** – You can define policies directly on the S3 bucket itself.
- **Access Control List (ACL)** – Sets policies to both a bucket and an object. With identity-based and resource-based policies, you cannot **assign a policy to an object**.

Note: You must provide public access to **both the bucket and object** in order to make it available to the world.

Read more about IAM in Amazon S3

S3 Storage Classes

A storage class represents the classification assigned to each object.

Each storage class has varying attributes that dictate:

- Storage cost
- Object availability
- Object durability
- Frequency of access

Each object must be assigned a storage class (standard is the default class)

Read more about Amazon S3 Storage Classes

S3 Storage Classes

- **S3 Standard** - For mobile applications, content distribution. ($0.023 per GB)
- **S3 Standard - Infrequent Access (IA)** - For backups. ($0.0125 per GB)
- **S3 One Zone IA** - Secondary backup, data you can recreate. ($0.01 per GB)
- **Glacier** - For long-term archival storage. ($0.004 per GB)
- **Glacier Deep Archive** - For long-term archival storage.
- **S3 Intelligent-Tiering** - Automatically moves objects between two access tiers based on access patterns

Object Durability is the percent (%) over a one-year time period that a file stored in S3 **will not be lost**.
Object Availbility is the percent (%) over a one-year time period that a file stored in S3 **will be accessible**.

S3 static web hosting

You can host static websites on S3 built with e.g., React and Angular. Static means, you build once that generates JS, CSS, and HTML. Those generated files are not dynamically changed. The static websites can be dynamic by calling the backend APIs. Because you can hard code the API URLs.

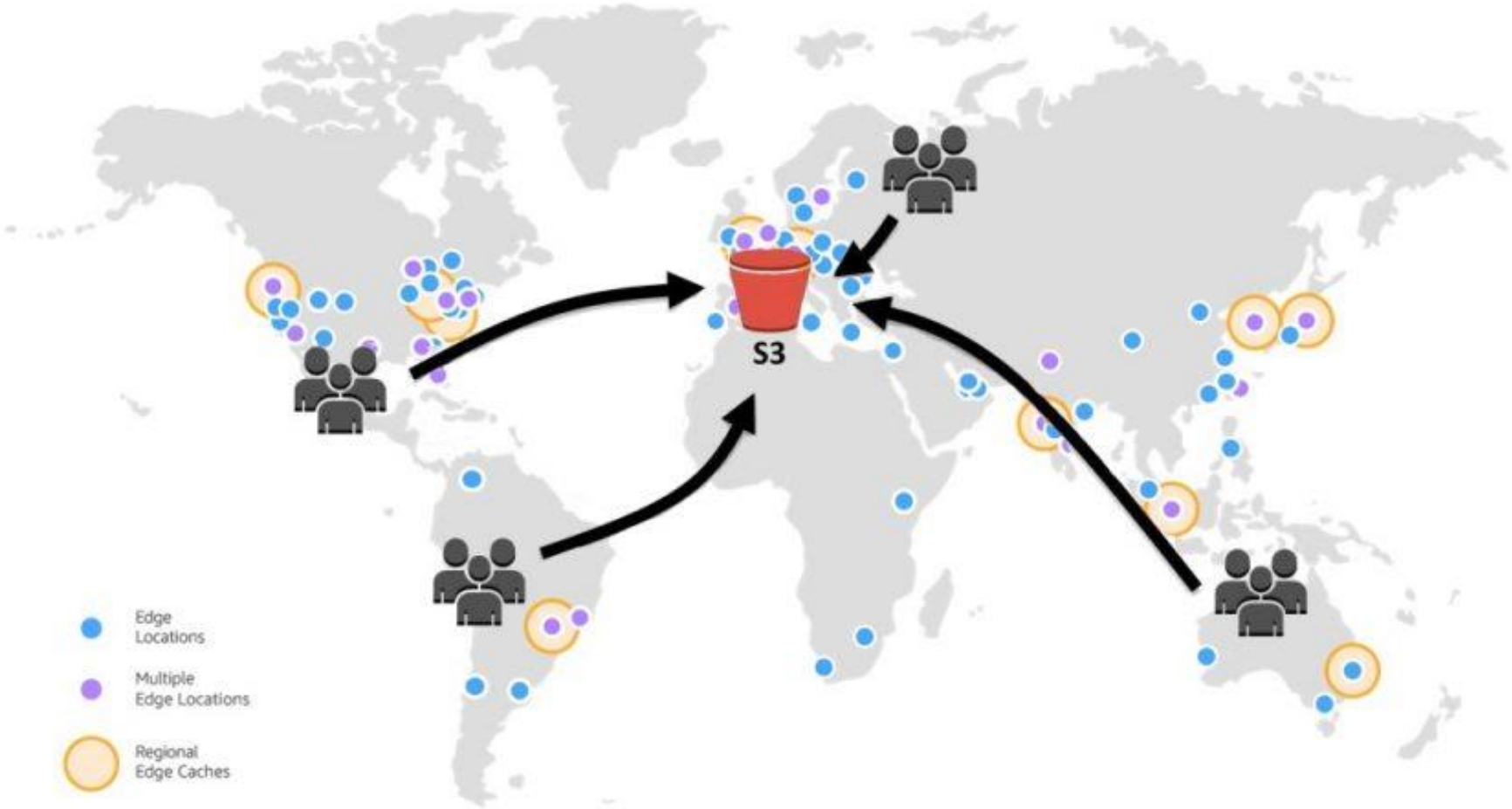
Static sites don’t require compute resources (server or CPU). You just have to distribute those files to clients. If you need to read something dynamically from the environment variable or disk, you would need a server.

Never store credentials and API keys in a static website. To log in to the static website, you can call the Login endpoint that returns JWT. And use the JWT for consecutive requests.

S3 Global CloudFront

- If you have a website, application, or another web resource, you probably have static content. Static content includes files like images, videos, or music, or even scripts like .css or js. In the pre-cloud era, you would put those files on a standard server, and then serve them on the internet to all of your viewers, across the globe, from one specific geo-location.
- Better Solution: Host your static web on S3 and distribute the content with AWS CloudFront which is a Content Delivery Network (CDN) service.
- Amazon CloudFront is a web service that **speeds up** distribution of your static and dynamic web content, such as .html, .css, .js, and image files, to your users. CloudFront delivers your content through a worldwide network of data centers called **edge locations**. When a user requests content that you're serving with CloudFront, the request is routed to the edge location that provides the **lowest latency** (time delay), so that content is delivered with the best possible performance.

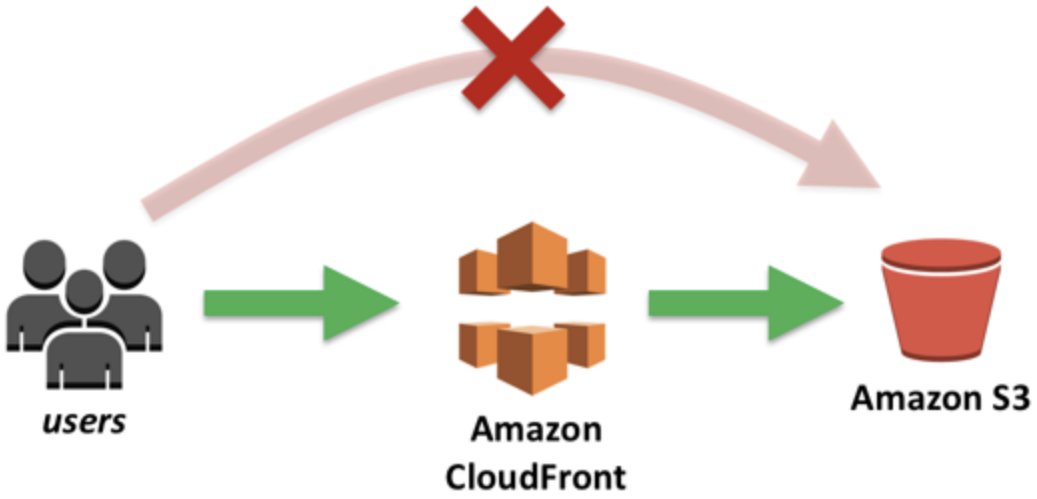
Static contents on S3 without CloudFront



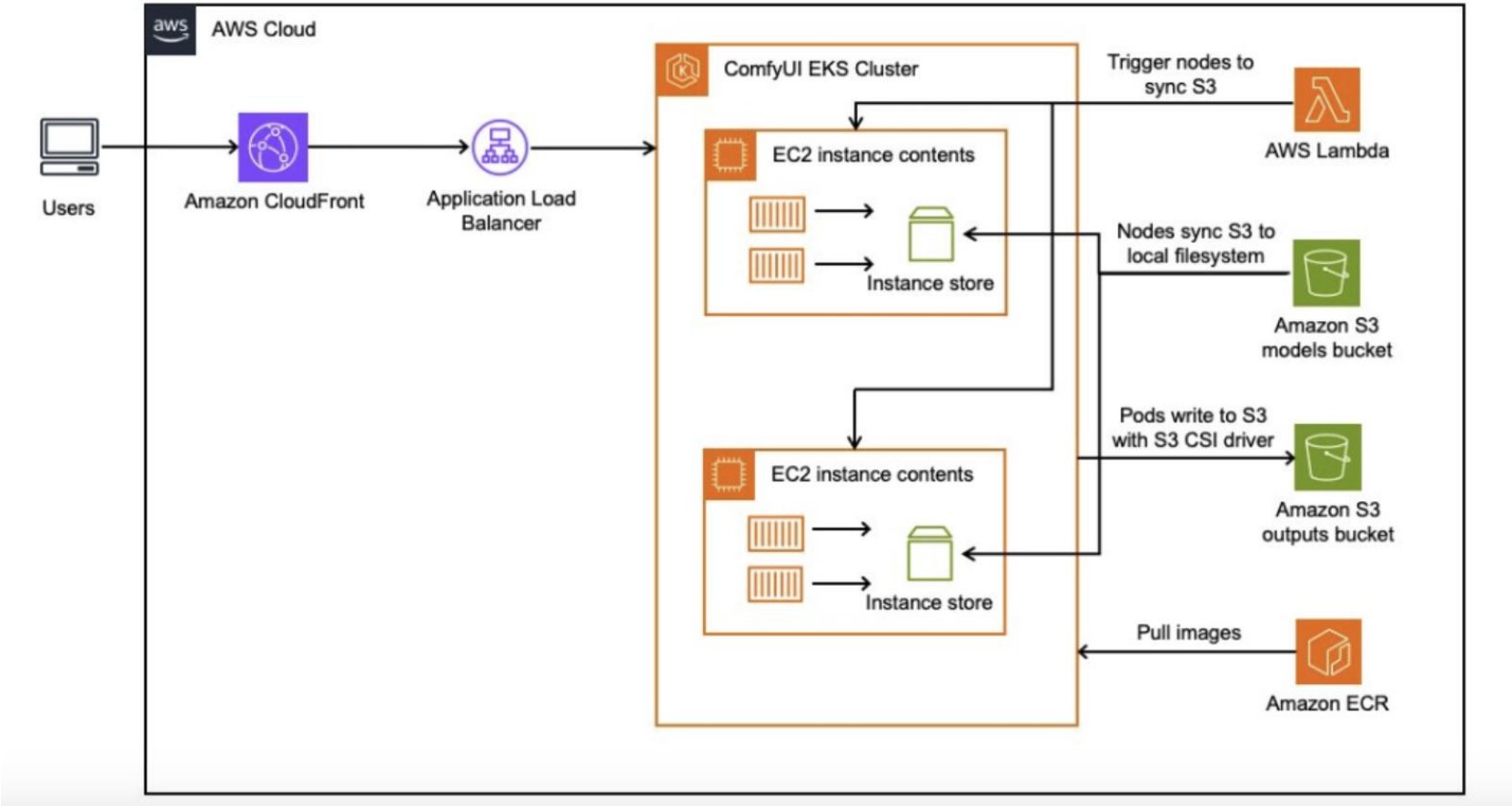
Static contents on S3 with CloudFront



Static contents on S3 with CloudFront



AWS CloudFront with Servers



Amazon CloudFront

- A CDN (Content Delivery Networks) that delivers content or API securely on AWS.
  - Frontend: Speeds up distribution of static and dynamic web content, such as .html, .css, .js, and image files.
  - Backend: Accelerate and secure the backend API.
- Features:
  - Edge Location Caching
  - HTTPS support
  - Integrate with AWS services like S3, EC2, ELB, and Lambda@Edge

Amazon CloudFront

- Content Distribution Process: The CloudFront delivers content by routing requests to the nearest edge location, which improves speed and reduces latency.
- Caching Mechanism: Data is cached across multiple global edge locations and the options available for invalidating or updating cached content.
- Security:
  - AWS Shield for DDoS
  - IAM for access control
  - SSL/TLS to encrypt transit data

Amazon CloudFront

- Integration of CloudFront with Route 53
  - Enhanced DNS Routing: Integrating Route 53 with CloudFront allows for DNS-level routing decisions that can direct users to the nearest CloudFront distribution, enhancing content delivery speed and reducing latency.
  - Using Route 53 Alias Records: To route traffic to a CloudFront distribution, which avoid DNS query charges and improves DNS resolution time compared to traditional CNAME records.
  - Health Checks and Failover: Route 53 health checks can monitor the health of CloudFront distributions and DNS failover mechanism can redirect users to alternative distributions if the primary one becomes unavailable.

S3 Pre-Signed URLs

- The S3-presigned URL stands out as a crucial feature, widely utilized in practical applications.
- A presigned URL uses security credentials to grant time-limited permission to download/upload objects from your applications.
- Use cases:
  - Secure file uploads
  - Temporary file downloads
  - Secure video streaming

Multipart Upload

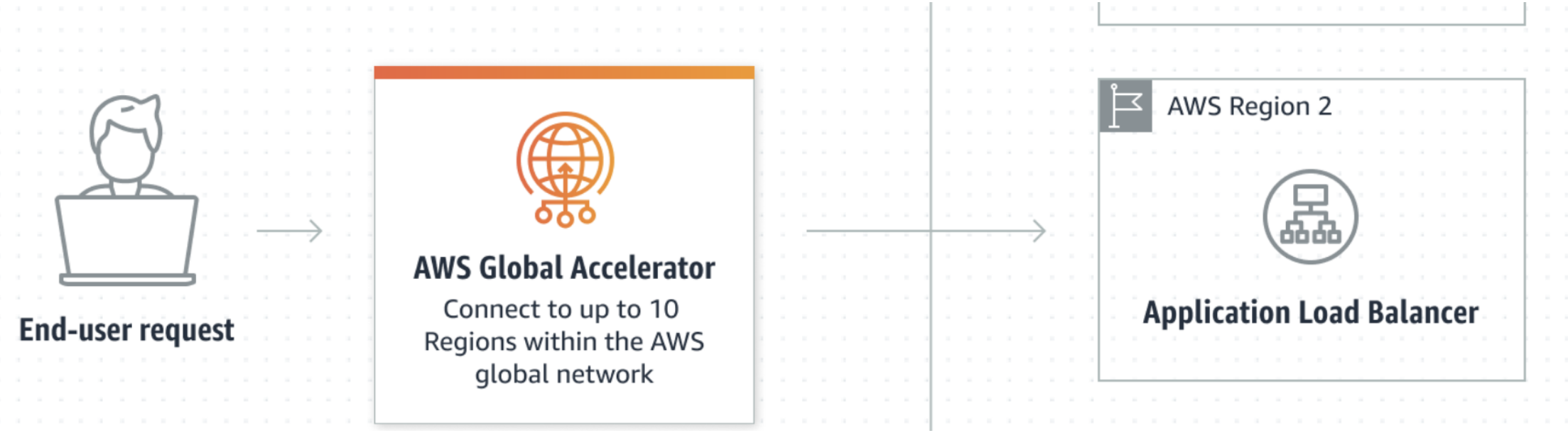
- Multipart upload allows you to upload a single object as a set of parts. Each part is a contiguous portion of the object's data. You can upload these object parts independently and in any order. If transmission of any part fails, you can retransmit that part without affecting other parts. After all parts of your object are uploaded, Amazon S3 assembles these parts and creates the object.
- In general, when your object size reaches **100 MB**, you should consider using multipart uploads instead of uploading the object in a single operation.
- Makes your app faster and complex. For example, more parameters to provide and edge cases such as what happens when there are incomplete files.

Multipart Upload benefits

- **Improved throughput** - You can upload parts in parallel to improve throughput.
- **Quick recovery from any network issues** - Smaller part size minimizes the impact of restarting a failed upload due to a network error.
- **Pause and resume object uploads** - You can upload object parts over time. After you initiate a multipart upload, there is no expiry; you must explicitly complete or stop the multipart upload.
- **Begin an upload before you know the final object size** - You can upload an object as you are creating it.

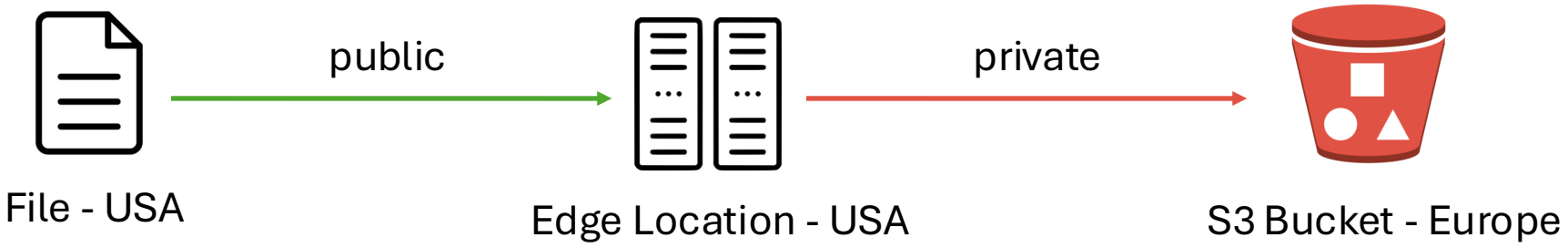
AWS Global Accelerator

- AWS Global Accelerator is a networking service that helps you improve the availability, performance, and security of your public applications



S3 Transfer Acceleration (upload only)

- Increase transfer speed by transferring file to an AWS edge location which will forward the data to the S3 bucket in the target region directly.
- To accelerate downloading, you can use CloudFront or S3 Global Replication.

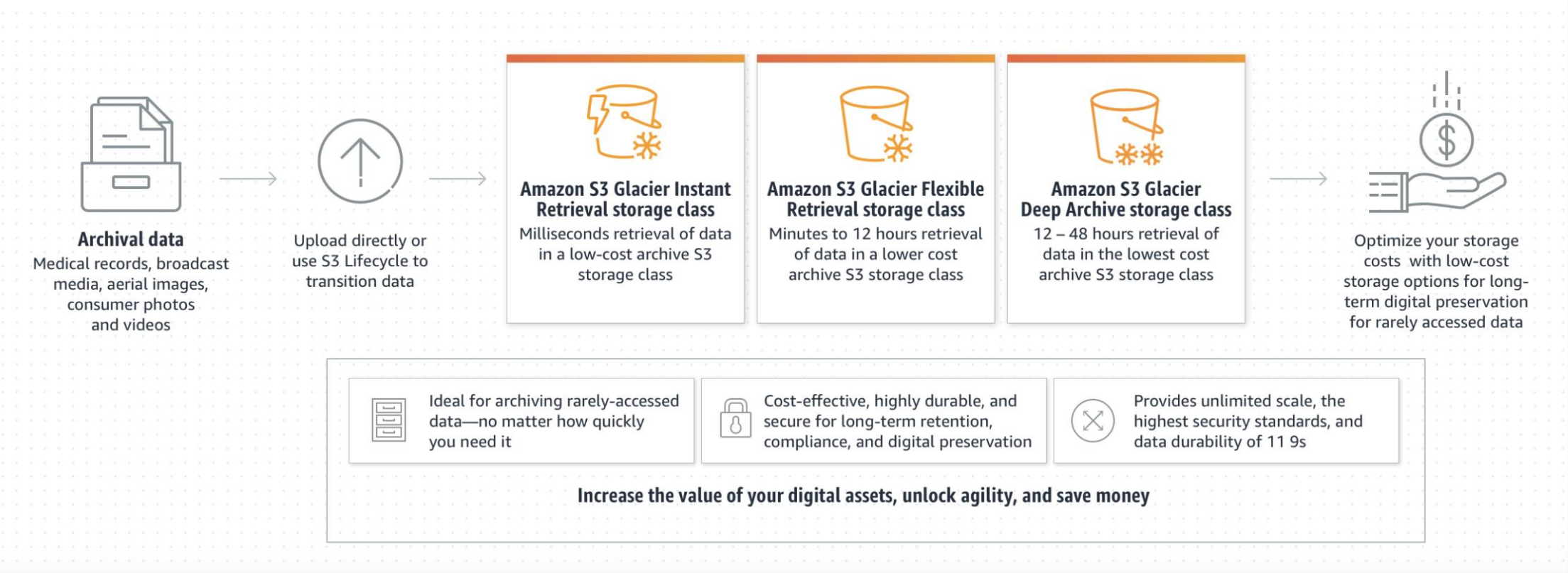


S3 Encryption

There are 4 methods of encrypting objects in S3:

- **SSE-S3**: Encrypts S3 objects using keys handled & managed by AWS. Out-of-box feature. Nothing to do on your side. All objects stored in S3 are encrypted.
- **SSE-KMS**: You can use AWS Key Management Service which provides keys to encrypt your data in S3.
- **SSE-C**: The key is generated by you (the appliance) and use that key to encrypt data.
- **Client-Side Encryption**: Encrypt before storing in S3 in your client app.

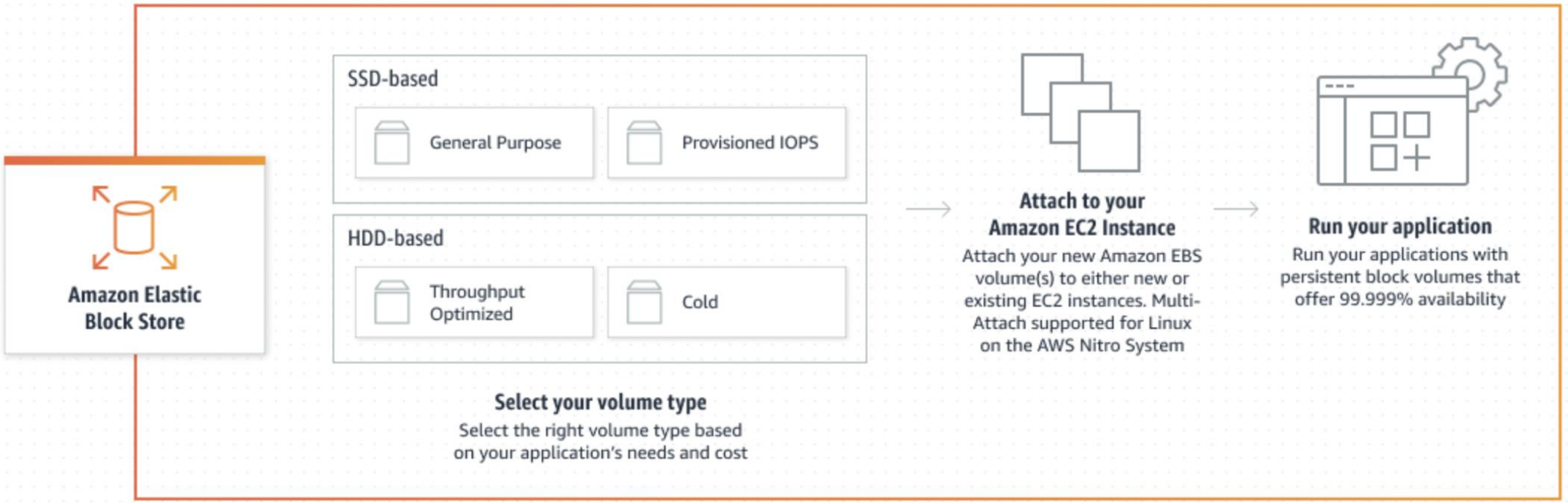
Amazon Glacier



Amazon Glacier

- A service for data archiving
- Performance
- Retrieval Flexibility
- Lowest cost for archiving storage

Amazon EBS


Amazon Elastic Block Store
SSD-based
General Purpose
Provisioned IOPS
HDD-based
Throughput Optimized
Cold
Select your volume type
Select the right volume type based on your application's needs and cost
Attach to your Amazon EC2 Instance
Attach your new Amazon EBS volume(s) to either new or existing EC2 instances. Multi-Attach supported for Linux on the AWS Nitro System
Run your application
Run your applications with persistent block volumes that offer 99.999% availability

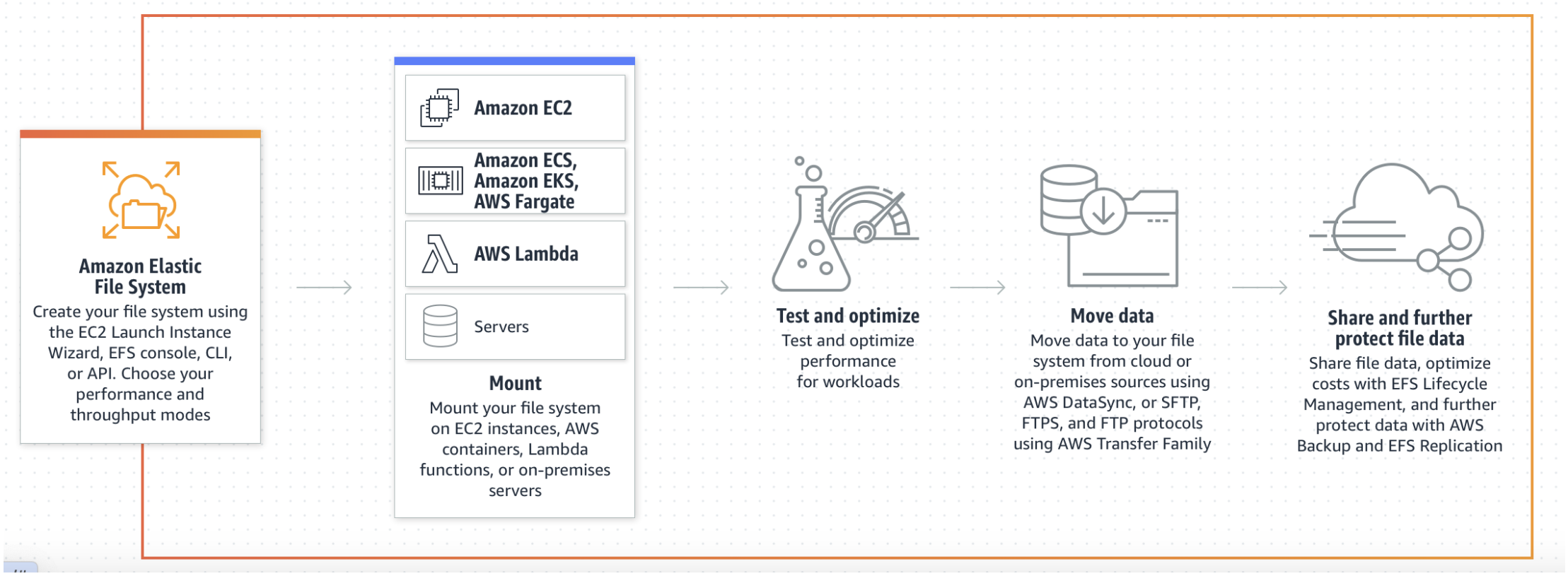
Amazon EBS

- EBS (Elastic Block Store) is a block-storage service designed for EC2 instances
- Durability and Availability
- Scale Fast
- High performance
- Optimize storage and cost
- Security

Amazon EBS

- Use cases:
  - Build mission-critical, I/O intensive applications in the cloud
  - Run Relational or NoSQL databases
  - Right-size big data analytics engines

Amazon EFS


Amazon Elastic File System
Create your file system using the EC2 Launch Instance Wizard, EFS console, CLI, or API. Choose your performance and throughput modes
Amazon EC2
Amazon ECS, Amazon EKS, AWS Fargate
AWS Lambda
Servers
Mount
Mount your file system on EC2 instances, AWS containers, Lambda functions, or on-premises servers
Test and optimize
Test and optimize performance for workloads
Move data
Move data to your file system from cloud or on-premises sources using AWS DataSync, or SFTP, FTPS, and FTP protocols using AWS Transfer Family
Share and further protect file data
Share file data, optimize costs with EFS Lifecycle Management, and further protect data with AWS Backup and EFS Replication

Amazon EFS

- EFS (Elastic File System): A managed NFS (Network File System) that provides file storage for use with other AWS services
- Scalability
- Performance
- Durability and Availability
- Security and Compliance

Amazon EFS

- Simplify DevOps
- Modernize application development
- Accelerate data science
- Enhance content management system

References

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- Google AI: https://gemini.google.com/app